Problem 6

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Problem 1 *Sketch a graph of the function f defined by*

$$f(x) = \begin{cases} 2x - 2\tan^{-1}(x) - x\ln(x^2+1) & \text{for } x \le 1 \\ \sqrt{x-2} & \text{for } x \ge 2 \end{cases}$$

(a) *Find the domain of f.*

(b) *Show that $f'(x) = -\ln(x^2+1)$ on the interval $(-\infty, 1)$.*

(c) *Compute a formula for f' on the interval $(2, \infty)$.*

(d) *Compute a formula for f'' on the interval $(-\infty, 1)$.*

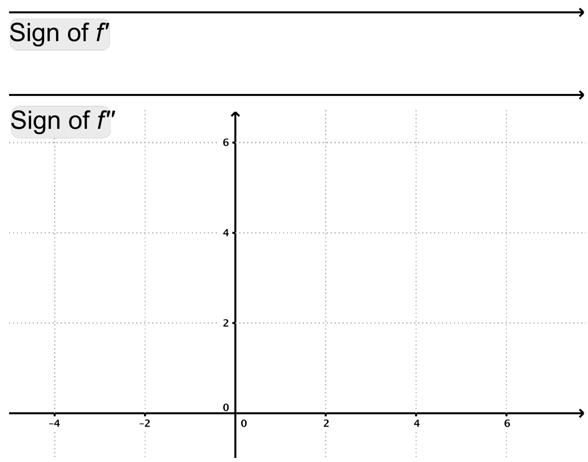
(e) *Compute a formula for f'' on the interval $(2, \infty)$.*

(f) *List all the asymptotes (with limit justifications, if appropriate). (If f does not have a particular asymptote write "NONE".)*

  (i) *Vertical asymptotes:*

  (ii) *Horizontal asymptotes:*

(g) *In the following sign chart for f', list the points (in the domain of f) where $f'(x) = 0$ or $f'(x)$ is undefined, indicate the sign of f' on each subinterval, and identify any local extrema (with "local max" or "local min".) In the following sign chart for f'', list the points (in the domain of f) where $f''(x) = 0$ or $f''(x)$ is undefined, indicate the sign of f'' on each subinterval, and identify any inflection points. Use this information to sketch a graph of f in the blank grid. (Be sure to indicate any x- or y-intercepts.)*


Sign of f'
Sign of f''
6
4
2
0
−4
−2
0
2
4
6